THE IMPACT OF AGRICULTURAL SUPPLY CHAIN DISRUPTIONS ON HEADLINE INFLATION IN MALAWI

BACHELOR OF SOCIAL SCIENCES IN ECONOMICS DISSERTATION

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BY

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DECLARATION

I, the undersigned, solemnly declare that this dissertation is the fruit of my own labour, original and true, and has not been submitted elsewhere for similar purposes. Wherever the work of others has been drawn upon, it has been duly acknowledged.

JUSTIN.A. MFAUME

Signature

Date

CERTIFICATE OF APPROVAL

The undersigned certify that this dissertation represents the student's own work and effort and has been submitted with our approval.

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Head of Department:

Signature: ………………………………Date: ……………………………………

Dean of Faculty:

Signature: ………………………………Date: ……………………………………

DEDICATION

I dedicate this paper to my loving and hardworking uncle, Mr. T. Malunga-my superhero, who fought bravely and took bullets for my success. You stood in the line of fire so that my path could be clear. Thank you for your unwavering love, wise guidance, and steadfast support throughout my journey as a son, a scholar, and a human being. Your courage and sacrifice have written themselves into who I am, and for this, I am forever grateful. May God bless you abundantly.

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ABSTRACT

This study investigates the impact of agricultural supply chain disruptions on headline inflation in Malawi, in light of persistent climatic shocks, input price volatility, and exchange rate fluctuations. The main objective is to assess how disruptions in the agricultural supply chain influence headline inflation in both the short and long run. Employing annual time-series data from 1970 to 2024, the study uses a Vector Error Correction Model (VECM) to capture the dynamic interactions between headline inflation and climatic shocks (proxied by annual rainfall), fertiliser prices, fuel prices, and exchange rate movements. Unit root and cointegration tests confirm a stable long-run relationship among the variables. The results indicate that climatic shocks, fuel prices, and exchange rate depreciation significantly drive headline inflation, while fertilizer prices are statistically insignificant, reflecting cushioning role of remittances, and government subsidies. The error correction coefficient of -0.48 implies that approximately 48% of deviations from long-run inflation equilibrium are corrected within one period. The study concludes that Malawi's inflation is largely supply-driven and highly sensitive to agricultural and external shocks. Policy recommendations include strengthening climate resilience, improving energy and transport infrastructure, and enhancing exchange rate management to ensure sustainable price stability in the agriculture-dependent economy.

Keywords: inflation, agricultural supply chain, climate shocks, fuel prices, exchange rate, vector error correction model.

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LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation/Acronym **Definition**

ADF	Augmented Dickey-Fuller
AIC	Akaike Information Criterion
AIP	Affordable Inputs Programme
AS	Aggregate Supply
CPI	Consumer Price Index
FAO	Food and Agriculture Organization of the United Nations
GDP	Gross Domestic Product
I(0)	Integrated of order zero (stationary at level)
I(1)	Integrated of order one (stationary after first differencing)
IFAD	International Fund for Agricultural Development
IMF	International Monetary Fund
IRFs	Impulse Response Functions
JB	Jarque-Bera statistic
LRAS	Long-Run Aggregate Supply
MERA	Malawi Energy Regulatory Authority
MWK	Malawian Kwacha
RBM	Reserve Bank of Malawi
SDGs	Sustainable Development Goals
SSA	Sub-Saharan Africa
TAM	Transporters Association of Malawi
UNW-DPC	United Nations World Data Protection Convention
USD	United States Dollar
USDA	United States Department of Agriculture
VECM	Vector Error Correction Model
WDI	World Development Indicators

CHAPTER ONE
INTRODUCTION

1.1 Background

Price stability is a cornerstone of sound macroeconomic management, supporting sustainable economic growth and the achievement of SDGs 1 (No Poverty) and 10 (Reduced Inequalities) (Mankiw, 2021). Central banks increasingly pursue this objective through inflation targeting, whereby a publicly announced medium-term inflation objective guides expectations and the use of policy instruments such as interest rates to maintain price stability. Stable prices protect real incomes, reduce uncertainty, and encourage investment and long-term planning (Gordon, 2012).

Inflation, defined as a sustained increase in the aggregate price level for most goods and services, can stimulate investment when moderate (3-6%), however at excessive double-digit rates it functions as a regressive tax (Gordon, 2012). Disproportionately harming low-income groups who rely on cash which loses value during inflationary periods, while wealthier individuals hold inflation-resistant assets like real estate and stocks, widening income inequality (McKinsey, 2024). Global income inequality, though declining since its 1980-2000 peak, after the 2008 financial crisis remains stark. The top 10% earn 52% of income, while the bottom 50% earn only 8% (World Inequality Lab, 2022). Regional disparities are notable, with Europe exhibiting greater equality and regions like the Middle East, Latin America, and Africa experiencing high inequality.

Severe inflation has profound socioeconomic consequences, as seen in Weimar Germany where bread prices soared from 0.26 to 80 billion Marks in 1923, and Zimbabwe's 2008 hyperinflation exceeding 79.6 billion percent monthly. These crises led to currency failures, adoption of foreign currencies, widespread suffering, income disparities, and violent social unrest (Jones, 2018).

Headline inflation includes both food and non-food components (Mankiw, 2021). Food inflation, the rate of price increases for CPI food items often drives headline inflation in agrarian and lower income economies where food dominates household spending (FAO, 2020). In March 2022, global food inflation surged to 9.93%, significantly outpacing non-food inflation at 7.41%, driven in part by widespread supply chain disruptions. (World Bank, 2023). Supply chains which refers to networks for producing and delivering goods, face interruptions from unexpected events like natural disasters, economic shocks, geopolitical conflicts, or health crises (Chopra & Meindl, 2016). Such disruptions cause reduced supply, higher costs, price volatility, and food insecurity.

Global agricultural supply chains face significant disruptions from extreme weather, geopolitical conflicts like the Russia-Ukraine crisis, and events such as the COVID-19 pandemic, driving inflation by reducing output and increasing input costs. In the 2010s, disruptions led to a 1.93% annual decline in agricultural output, directly spiking consumer prices (USD, 2022). Fertilizer costs surged by 80% in 2021 and an additional 30% by early 2022, escalating production expenses and sustaining inflationary pressures (Jenkins, 2023).

Sub-Saharan Africa (SSA) is highly vulnerable to inflation due to climate-induced agricultural disruptions and structural constraints. Climate shocks such as East Africa's droughts reduced yields by 10-15%, contributing to 17.2% food inflation in 2022 (Tankari & Fofana, 2024). Weak transport and irrigation infrastructure, combined with global commodity price spikes and currency depreciation, raised input costs and pushed SSA's headline inflation to 14.5%, well above the global average of 8.7% (World Bank, 2023).

Malawi's economy is highly fragile due to its reliance on rain-fed agriculture, which contributes 30% of GDP, employs 64% of the workforce, and generates over 80% of export earnings, primarily from tobacco (RBM, 2024). For many Malawians, inflation is not an abstract statistic; it manifests in overnight maize price surges that directly affect the cost of feeding a family, erode purchasing power, and trap households in poverty. Climate shocks, including Cyclone Freddy in 2023, which reduced maize output by 30%, and recurring droughts that lower yields by 15-20% annually, threaten food security and export earnings (IFAD, 2023). Rising costs of imported fertilizer up 200% from 2020 to 2024 combined with fuel price hikes (44.9% for diesel in June 2022) and a 44.4% devaluation of the Kwacha, drove headline inflation to 34.5%, with food inflation at 43.5% compared to 22.8% for non-food items by December 2023 (Nation Online, 2024).

Malawi's Southern Region was hardest hit by the 2024 El Niño drought and Cyclone Freddy in 2023, causing severe maize crop losses and leaving 2.46 million people in need of aid (Self Help Africa, 2024). As maize constitutes 53.7% of Malawi's CPI food basket, these regional supply shocks significantly contributed to national inflation. In June 2024, Southern Region maize prices averaged K1,067/kg, exceeding the Central and Northern region prices of K1,036/kg and K913/kg, respectively, with spot checks reaching K1,300/kg (Nation Online, 2024).

1.2 Problem Statement

Malawi faces chronic headline inflation averaging 21% since the 1990s and surging to 34.5% in 2023, disproportionately driven by food inflation (43.5%) consistently double non-food rates due to structural supply-side vulnerabilities, including climatic shocks disrupting maize production, surging global input costs (e.g. fertilizer and fuel), and exchange rate volatility amplifying imported inflation (Chirwa, 2023). Despite interventions like the Reserve Bank of Malawi's demand-side monetary tightening and the government's Affordable Inputs Programme (AIP), excessive inflation persists, threatening macroeconomic stability and food security, as policies inadequately address supply-chain disruptions, climate volatility, and input price shocks (Chavula).

Prior studies, including Simwaka (2018) and Khonje (2020), emphasize demand-side drivers like money supply growth, while Kalua (2017) examines the isolated effect of agricultural GDP on inflation, overlooking the combined impact of interconnected supply-side shocks. This study quantifies their collective role in driving inflation and evaluates why Malawi's current policies fail to effectively address inflationary pressures allowing policy refinements to better guide ant inflation efforts.

1.3 Research Objectives

1.3.1 Main Objective

To investigate the impact of agricultural supply chain disruptions on headline inflation in Malawi.

1.3.2 Specific Objectives

i. To analyze the correlation between annual rainfall variations and headline inflation in Malawi.

ii. To quantify the contribution of fertilizer price dynamics on headline inflation in Malawi.

iii. To assess the transmission effects of fuel price shocks on headline inflation in Malawi.

1.4 Study Hypothesis

From the above stated specific objectives, the following null hypotheses will be tested;

i. No significant relationship between annual rainfall and headline inflation in Malawi
ii. Fertilizer price dynamics do not significantly contribute to food inflation in Malawi.
iii. Fuel price changes do not significantly contribute to headline inflation in Malawi.

1.5 Research Rationale and Justification

This study focuses on Malawi, an agriculture-dependent economy where food prices especially maize play a major role in headline inflation. The country's reliance on rain-fed agriculture, road transport and imported farm inputs makes inflation highly sensitive to supply chain disruptions. The study period allows examination of both short-run shocks and long-run inflation trends. A Vector Error Correction Model (VECM) analyses the short-run and long-run effects of annual rainfall variation, fertilizer prices, fuel price shocks, and exchange rate on headline inflation. The study is timely given recent inflationary pressures from climate shocks, rising input costs, and exchange rate volatility.

1.6 Significance of the Study

This research advances academia and policy by aligning with Malawi Vision 2063’s pillars of agricultural productivity, commercialization, and inclusive wealth creation, promoting sustainable economic growth through effective management of stable, low inflation. It supports SDGs, including Zero Hunger (SDG 2), Decent Work and Economic Growth (SDG 8), and Climate Action (SDG 13), by enhancing resilience to shocks. Academically, it enriches the literature with a Malawi-specific model integrating multiple supply-side shocks on inflation, providing a replicable framework for agro-dependent economies and linking cost-push inflation theories to supply chain dynamics. For policy, it offers evidence-based recommendations for stakeholders, such as the Reserve Bank of Malawi and the Ministry of Agriculture, addressing structural vulnerabilities to mitigate inflation, enhance food security, and foster economic resilience.

1.7 Organization of the Study

The rest of the study is organized into four chapters. Chapter Two reviews relevant theoretical and empirical literature on agricultural supply chain disruptions and inflation. Chapter Three outlines the methodology, including research design, data sources, model specification, variable descriptions, and diagnostic tests. Chapter Four presents and discusses the empirical results from the econometric analysis. Chapter Five concludes by summarizing key findings, offering policy recommendations, and suggesting areas for further research.

CHAPTER TWO
LITERATURE REVIEW

2.1 Introduction

This chapter reviews literature on the impact of agricultural supply chain disruptions on inflation in Malawi. It comprises a theoretical review of economic theories and supply chain disruption frameworks, and an empirical review of pertinent studies. The chapter ends by identifying the research gap, underscoring this study's unique contribution.

2.2 Theoretical Review

This section examines the theoretical foundations explaining how disruptions in agricultural supply chains affect price levels.

2.2.1 Cost-Push Inflation

Cost-push inflation refers to a rise in the overall price level resulting from increased production costs. This theory gained prominence in the mid-20th century as an alternative to demand-pull inflation, particularly after economists observed stagflation, simultaneous inflation and low economic growth with high unemployment. An important early study by A. W. Phillips in 1958 highlighted an inverse relationship between unemployment and wage inflation, suggesting that rising labor costs can drive price increases. The theory's importance was underscored during the 1970s oil crises, when sharp increases in oil prices, a critical input, were passed directly to consumers, triggering inflation despite weak aggregate demand (Gordon, 2012).

The aggregate supply (AS) framework provides a useful structure for analyzing cost-push inflation. Aggregate supply describes the relationship between output and inflation and is distinguished between the short run and the long run. In the long run, wages and prices are fully flexible, and the long-run aggregate supply (LRAS) curve is determined by real factors such as labour, capital, technology, and the natural rate of unemployment (Mishkin, 2012). Since these factors are assumed to be independent of inflation, the LRAS curve is vertical at the level of potential output, Y_P, defined as the sustainable level of output at which all factors of production are efficiently utilized and output is maximized without generating inflation. Any deviation from this level induces changes in inflation that eventually restore output to its potential.

In the short run, nominal rigidities (downward stickiness) prevent wages and prices from adjusting instantaneously. As a result, output may deviate from its potential level. When output rises above potential, tight labour and product markets exert upward pressure on inflation. Consequently, the short-run aggregate supply (AS) curve is upward sloping: higher output relative to potential is associated with higher inflation (Mishkin, 2012). A negative supply shock, defined as an unexpected event that reduces the supply of goods and services, shifts the short-run AS curve leftward (or upward), leading to higher inflation and lower output. If such shocks are temporary, the economy eventually returns to its original potential output, but with a higher inflation rate in the short run. In contrast, permanent negative supply shocks reduce the economy's productive capacity, shifting the LRAS curve leftward and resulting in a permanently higher inflation rate.

These mechanisms are illustrated in Figures 1 and 2.

Figure 1

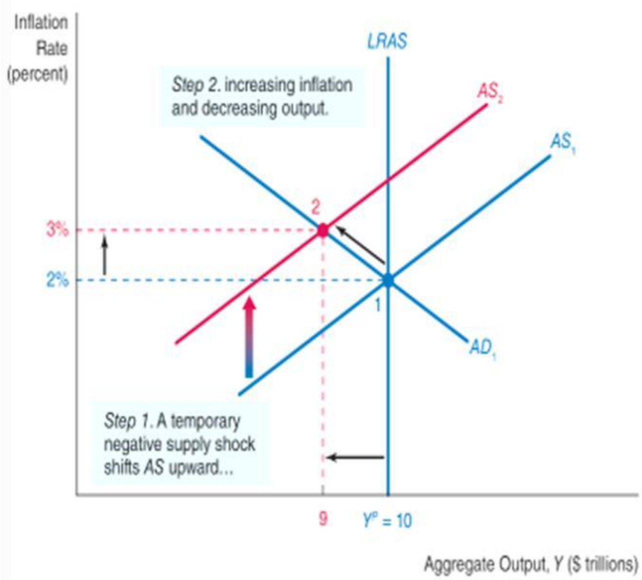
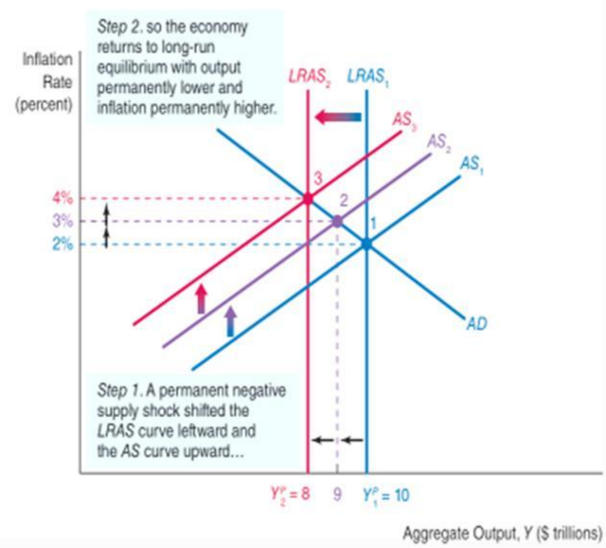


Figure 2



Source: F. Mishkin (2012) *Macroeconomics: Policy and Practice.*

Figure 1 depicts a temporary negative supply shock that shifts the short-run AS curve from AS_1 to AS_2, raising inflation from, for say, 2% to 3% while reducing output below potential. Figure 2 shows a permanent negative supply shock that shifts both the short-run AS and LRAS curves leftward from $LRAS_1$ to $LRAS_2$, lowering potential output and increasing inflation permanently, for instance from 3% to 4% (Mishkin, 2012).

In Malawi, cost-push inflation is closely linked to climate-related shocks, exchange-rate depreciation, and rising fuel prices. Climate shocks such as droughts and floods reduce agricultural output, leading to lower exports and shortages of key food commodities. As staple foods such as maize become scarce, and inputs for cooking oil production such as groundnuts, sunflower, and soya beans decline, food prices rise sharply, exerting upward pressure on headline inflation. These supply-side disturbances are particularly inflationary given the high weight of food in Malawi's consumer price index.

Exchange-rate depreciation further amplifies cost-push pressures. When the kwacha depreciates against the US dollar, the cost of imported agricultural inputs rises. Higher input prices increase production costs for farmers, discouraging the adoption of capital-intensive technologies such as advanced irrigation systems and contributing to persistently low output. The resulting increase in farm-gate prices is transmitted to consumer prices, reflecting imported inflation. This channel is especially important in Malawi, where agricultural production depends heavily on imported inputs.

Rising global oil prices, combined with exchange-rate depreciation, increase domestic fuel prices particularly diesel which is critical for transportation and production. In addition, logistical disruptions, such as conflicts along key transport routes or strikes by the Transporters Association of Malawi (TAM), can lead to fuel shortages. Under such conditions, fuel may only be available on informal markets at prices substantially above official levels. Higher transport costs raise the cost of moving agricultural inputs to farms and outputs to markets, and these costs are ultimately passed on to consumers in the form of higher prices, further intensifying cost-push inflation.

2.2.2 Structuralist Inflation Theory

Structuralist inflation theory posits that inflation in developing countries arises from entrenched structural rigidities that limit supply-side responsiveness to demand changes, rather than purely monetary causes. Formulated mainly by Latin American economists such as Raúl Prebisch and the Economic Commission for Latin America and the Caribbean ECLAC during the 1950s-60s, the theory explains persistent high inflation despite conventional anti-inflation policies (Prebisch, 1950). These scholars identified bottlenecks in agriculture, underdeveloped infrastructure, and over-reliance on a few primary commodities as creating inelastic supply, leaving economies vulnerable to price shocks (Dresdner, 1990).

Malawi exhibits similar structural constraints exacerbated by agricultural supply chain disruptions. Poor road networks, limited irrigation, and inadequate storage amplify inflationary effects from rising fuel prices or drought-induced crop losses. The heavy dependence on rain-fed agriculture increases vulnerability to weather and climatic shocks such as droughts. These systemic weaknesses cause even minor shocks to disproportionately affect prices, as the economy cannot effectively adjust (World Bank, 2020).

2.2.3 Demand-Pull Inflation

Demand-pull inflation occurs when aggregate demand exceeds aggregate supply, leading to an increase in the general price level. Grounded in Keynesian economics, this theory was introduced by John Maynard Keynes in his 1936 work "The General Theory of Employment, Interest and Money" where he explained that when an economy nears full employment, fiscal or monetary expansion leads to price increases as production capacity limits are reached (McEachern, 2006).

Increases in money supply enhance purchasing power, while GDP growth indicates rising aggregate demand. Though agricultural supply chain disruptions primarily generate cost-push pressures, they may indirectly influence demand; for example, climatic crop failures may reduce farmer incomes and demand, whereas government subsidies could boost it, consequently triggering a demand shock. Hence, capturing demand-side variables is necessary to fully understand headline inflation and isolate supply shock effects (McEachern, 2006).

2.2.4 Networking Theory of Supply Chain Disruption

Networking theory views supply chains as complex, interconnected systems where disruptions in one part propagate throughout the network, often amplifying impacts through ripple effects. This perspective, which gained prominence in the early 21st century, treats the supply chain as a network of nodes (entities such as suppliers, manufacturers, distributors, and customers) and links (flows of materials, information, and funds). Scholars like Yossi Sheffi emphasized this approach in supply chain risk management, highlighting how vulnerabilities arise from interdependencies and limited redundancy (Sheffi, 2005).

Figure 3: Multi-tier supply chain with upstream and downstream flows

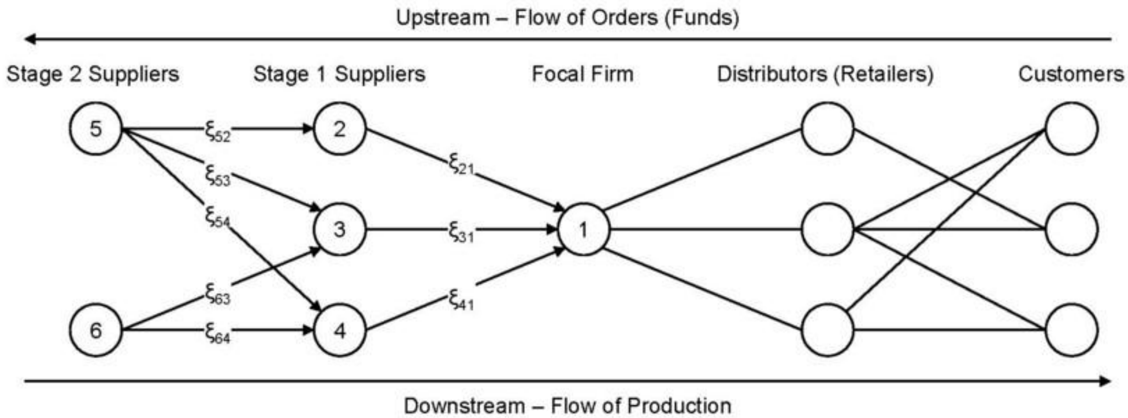


Figure3 illustrates a supply chain as a network made up of several connected stages, with a central firm such as a manufacturer at the core of the system, which organizes production by placing orders and sending payments upstream to its direct suppliers (Stage 1 suppliers), who in turn depend on other suppliers (Stage 2 suppliers) for raw materials and basic inputs, while downstream the finished goods produced by the central firm move to distributors or retailers and ultimately reach final consumers, clearly showing that each stage in the chain is linked to and depends on the others, so that a disruption such as a delay, shortage, or failure at an early supplier can spread through the network, interrupt production at the central firm, and eventually affect the availability and prices of goods faced by consumers.

2.3 Empirical Review

Empirical literature provides crucial insights into the relationship between agricultural supply chain disruptions and inflation, particularly within agrarian economies such as Malawi.

A substantial body of research from Sub-Saharan Africa (SSA) highlights how climatic variability, volatility in input prices, and exchange rate fluctuations intensify inflationary pressures. These studies reveal the complex interactions between supply-side shocks and macroeconomic outcomes, with significant policy implications for countries like Malawi, where agriculture constitutes a large share of GDP and employment. This section reviews key empirical findings from Malawi and the broader SSA region, focusing on how agricultural supply chain disruptions influence headline inflation through their effects on production and distribution channels.

In Malawi, several studies consistently link agricultural disruptions to inflationary outcomes, reflecting the sector's central role in the economy. Kalua (2017) employed a Vector Autoregressive (VAR) model using data from 1985 to 2015 to examine agricultural determinants of inflation. The results indicated a pronounced inverse relationship, where a 1% increase in agricultural GDP, tobacco production, and tobacco prices reduced inflation by 31%, 11%, and 11% respectively. This underscores the stabilising influence of agricultural productivity on price stability. Similarly, Chirwa and Chinsinga (2011) reported a 141% real rise in staple food prices, particularly maize, between 1998 and 2008, attributing this surge mainly to supply constraints within the agricultural sector. These price increases severely affected household welfare and macroeconomic stability.

Recent reports by the Reserve Bank of Malawi (2023) reaffirm this pattern, noting that Malawi's inflation rates remain among the highest in SSA, driven largely by food price volatility and agricultural bottlenecks. The UN World Disaster Prevention Committee (UNW-DPC, 2008) further documented how recurrent droughts have sharply reduced maize output, heightening food insecurity and inflationary pressures. Complementing these findings, Nkomboni and Ayal (2022) established that climate variability significantly increases price instability, particularly in economies dependent on rain-fed agriculture such as Malawi. The Shamba Centre (2025) also linked the 2023 maize harvest shortfall to rising fertilizer costs in 2022, which curtailed fertilizer use and worsened food insecurity.

Mangani (2013) offers a contrasting perspective by challenging conventional fiscal dominance and demand-pull explanations of inflation in Malawi. Instead, his findings highlight external factors particularly exchange rate depreciation and imported inflation as the dominant sources of price instability. The evidence indicates that fiscal deficits and money supply growth exert limited influence on inflation, thereby underscoring the critical role of trade openness and exchange rate volatility in driving inflation dynamics through higher landing costs of agricultural inputs. Supporting this argument, World Bank (2016) estimates reveal that every 10% increase in fuel prices leads to a 1.2% rise in Malawi's inflation rate.

Furthermore, the Reserve Bank of Malawi (2024) further reported that exchange rate fluctuations account for approximately 30-40% of inflation volatility, highlighting the substantial influence of external shocks on domestic price stability.

Across the wider SSA region, climatic shocks persist as a recurrent economic challenge, exacerbating supply chain weaknesses and fueling inflation. Tankari and Fofana (2024), using Seasonal Autoregressive Integrated Moving Average (SARIMA) models, demonstrated that climatic shocks substantially reduce agricultural yields, leading to food shortages and steep price increases across African economies. Their conclusions align with Kunawotor et al. (2022), who observed similar dynamics across SSA.

Dawe (2008) investigated the vulnerability of developing nations to oil price fluctuations, revealing that dependence on rain-fed agriculture and weak infrastructure amplify the transmission of fuel price rises into higher consumer prices. In the same vein, Adeniyi and Yusuf (2023) identified a statistically significant positive relationship between oil price shocks and headline inflation in Nigeria a pattern that reflects broader regional experiences.

2.4 Modelling Agricultural Supply Chain Impacts on Inflation

Prior studies have used various econometric approaches to analyse inflation dynamics. In Malawi, Kalua (2017) applied a VAR model, while Mangani (2013) employed SVAR and ARDL frameworks. At the regional level, Tankari and Fofana (2024) used SARIMA models to assess the inflationary effects of climatic shocks in Sub-Saharan Africa. While informative, these approaches have limitations: VAR, SVAR, and SARIMA require variables to be stationary and are weak in capturing long-run relationships, and SARIMA depends on high-frequency data often unavailable for key agricultural variables. Although ARDL allows mixed integration orders, it is prone to endogeneity and spurious regression concerns.

To address these limitations, this study adopts a Vector Error Correction Model (VECM), which is appropriate for cointegrated variables and enables simultaneous analysis of short-run dynamics and long-run equilibrium relationships. The inclusion of an error correction term captures the speed of adjustment following shocks, providing a robust framework for assessing how agricultural supply chain disruptions and macroeconomic factors jointly influence inflation in Malawi over 1970-2024.

2.5 Research Gap

Existing literature offers insights into agricultural shocks and inflation but lacks a comprehensive analysis of their combined effects in Malawi. Studies like Kalua (2017) focus on agricultural productivity, while Tankari and Fofana (2024), UNW-DPC (2008), Dawe (2008), and Adeniyi and Yusuf (2023) examine isolated factors such as climatic shocks or oil price volatility. However, they fail to quantify the simultaneous impact of multiple shocks drought-related maize declines, rising fertilizer costs, and currency fluctuations on headline inflation through Malawi's agricultural supply chain. Unlike prior studies, this research employs a Vector Error Correction Model (VECM) to analyze both short-run and long-run dynamics of cointegrated variables, providing a holistic framework to inform targeted policy interventions for Malawi's agrarian, import-dependent economy.

CHAPTER THREE
METHEDOLOGY

3.1 Introduction

This chapter outlines the methods used to collect and analyze secondary data, guided by the study objectives, the nature of the research, and relevant theoretical and empirical literature on the relationship between supply chain disruptions and headline inflation.

3.2 Sources of Data

The study employed secondary annual time-series data for Malawi spanning the period 1970-2024. The dataset was compiled from reputable national and international institutions to ensure reliability, consistency, and adherence to internationally accepted statistical standards.

Money supply and headline inflation data were obtained from the Reserve Bank of Malawi (RBM). As the country's official monetary authority, RBM compiles and disseminates macroeconomic statistics in line with international accounting and statistical frameworks, making its data on monetary aggregates and inflation authoritative and widely used in macroeconomic research.

Exchange rate data were sourced from the World Bank's World Development Indicators (WDI), which harmonize national data from credible institutions such as central banks and national statistics offices and ensure cross-country comparability through standardized methodologies.

Annual rainfall variation data were collected from NASA's Power Earth Observations database, which provides standardized, satellite-derived climate datasets validated with ground-based meteorological records, making them suitable for long-term climatic analysis.

Fuel price data were obtained from the Malawi Energy Regulatory Authority (MERA), the statutory body mandated under the Energy Regulation Act of 2004 to regulate energy pricing and standards. MERA's fuel price data are based on import landing costs, global oil price movements, and regulated tax structures, ensuring accuracy and validity.

Fertilizer price data were obtained from the World Bank Commodity Price Data and the United States Department of Agriculture (USDA) website, which provide internationally benchmarked agricultural input price information based on global market conditions, trade data, and production costs.

Annual Rainfall was represented by national annual average rainfall, serving as a proxy for weather and climatic shocks at a countrywide level rather than district or regional differences. This methodological consistency enhances comparability across variables and strengthens the robustness of empirical findings.

Annual data were used exclusively to ensure temporal uniformity, mitigate inconsistencies arising from differing reporting frequencies, and focus on the long-run inflationary relationships central to the study. Variables such as inflation, money supply, and exchange rates are often reported at higher frequencies (monthly or quarterly), while rainfall and fuel prices exhibit seasonal or irregular patterns.

3.3 Model Specification

To examine the impact of agricultural supply chain disruptions on headline inflation in Malawi, this study employs a Vector Error Correction Model (VECM), ideal for analyzing dynamic and cointegrated relationships among time-series variables. The VECM captures both short-run and long-run effects, enabling a comprehensive assessment of how disruptions influence inflation. The model includes headline inflation (INF) as the dependent variable and independent variables: annual rainfall amounts (RainVar) as a proxy for climatic shocks, urea fertilizer prices (FertP), diesel prices (FuelP) as a proxy for oil price shocks, exchange rate (ExR), and broad money supply (MS).

The model is specified to reflect the main objective of the study, with headline inflation as the dependent variable and a set of independent variables representing agricultural supply chain disruptions. The general model as adopted from Adeniji, S. (2013) can be expressed as:

$\mathrm{INF}_t = f$ **(**MS, RainVar, FertP, FuelP, ExR**)**

$$\begin{aligned}\Delta\mathrm{INF}_t &= \alpha + \sum_{i=1}^{p} \beta_i \Delta\mathrm{INF}_{t-i} + \sum_{i=1}^{p} \gamma_i \Delta\mathrm{MS}_{t-i} + \sum_{i=1}^{p} \delta_i \Delta\mathrm{RainVar}_{t-i} \\ &+ \sum_{i=1}^{p} \varepsilon_i \Delta\mathrm{FertP}_{t-i} + \sum_{i=1}^{p} \zeta_i \Delta\mathrm{FuelP}_{t-i} + \sum_{i=1}^{p} \eta_i \Delta\mathrm{ExR}_{t-i} + \theta\mathrm{ECM}_{t-1} + e_t\end{aligned}$$

Where:

- INF_t = Headline Inflation in Malawi at time t
- MS = Money Supply
- RainVar = Annual Rainfall
- FertP = Fertilizer Prices
- FuelP = Fuel Prices
- ExR = Exchange Rate
- Δ = First-difference operator
- ECM_{t-1} = Error Correction Term, capturing the long-run relationship
- $\alpha, \beta, \gamma, \delta, \lambda, \rho, \zeta, \phi$ = Model parameters to be estimated
- ϵ_t = The error term

3.4 Variable Definition

3.4.1 Dependent Variable

Headline Inflation (INF)

Gordon (2012) defines headline inflation as the measure of the total inflation rate of an economy over a specific period, capturing changes in the prices of all goods and services, including volatile components such as food and energy. In this study, headline inflation is measured as the year-on year annual percentage change in the average Consumer Price Index (CPI) and reflects the overall movement in the general price level. By reflecting price changes in essential consumption items, headline inflation captures the fall in purchasing power actually experienced by households and businesses and serves as a key indicator for assessing inflationary pressures and guiding monetary policy and macroeconomic stability analysis.

3.4.2 Independent Variables

Annual Rainfall (RainVar)

Annual rainfall is the total amount of precipitation received in a country over a year and is commonly used as an indicator of climatic conditions affecting agricultural production. Variations in annual rainfall influence crop yields, food availability, and supply conditions, thereby affecting food prices and headline inflation, particularly in agriculture-dependent economies (NASA, 2020).

Fertilizer Prices (FertP)

Fertilizer refers to chemical or organic substances applied to soil or crops to supply essential nutrients and enhance agricultural productivity (USDA, 2022). In this study, fertilizer prices are measured using international urea prices, expressed in US dollars per metric ton, as urea is the most widely used nitrogen-based fertilizer.

Fuel Prices (FuelP)

The annual average diesel price was adopted as a proxy for fuel prices since diesel is predominantly used in operating agricultural machinery and transporting inputs and outputs, linking it directly to production and supply chain cost (Jenkins, 2023). A positive sign is expected; as increased fuel costs directly raise the price of moving goods from farms to markets.

Exchange Rate (ExR)

The exchange rate refers to the price of one country's currency in terms of another country's currency. In this study, it is measured as the value of the Malawi kwacha against the US dollar and captures the cost of imported goods and inputs. Movements in the exchange rate directly affect domestic prices in import-dependent economies, as currency depreciation raises the local currency cost of imports and contributes to imported inflation (Mishkin, 2012).

Money Supply (MS)

Money supply refers to the total stock of money available in an economy at a given point in time, including currency in circulation and various forms of deposits held by the public. It represents the primary liquidity available for transactions, savings, and investment and serves as a key indicator of aggregate demand pressures in the economy (Gordon, 2012).

Table 1: Table of Expected Signs

Variable	Parameter	Expected Sign	Justification	Authors
Lagged Inflation	β_1	+	inflation inertia	Cogley & Sargent (2005)
Annual Rainfall	β_2	-	Adequate rainfall eases headline inflation by reducing food-price pressures.	FAO (2020); Lobell et al. (2011)
Fertilizer Prices	β_3	+	Increase production costs (cost-push)	World Bank (2023), Jenkins (2023)
Fuel Prices	β_4	+	Increase transportation costs (cost-push)	Adeniyi & Yusuf (2023)
Exchange Rate	β_5	+	increased import costs	IMF (2021)
Money Supply	β_6	+	Reflects demand-pull inflation	(Gordon, 2012).

3.5 Pre-Estimation Diagnostic Tests

To ensure correct specification of the VECM results, several diagnostic tests will be conducted.

3.5.1 Unit Root Test

The Augmented Dickey-Fuller (ADF) test will be used to determine the stationarity of the time series data, a prerequisite for VECM analysis, which requires all variables to be integrated of the same order, typically I(1). It identifies the number of differences needed, ensuring all variables are stationary only after the first difference (Enders, 2014).

3.5.2 Cointegration Test

The study will employ the Johansen cointegration test to determine if a long-run relationship exists between the non-stationary variables and select number of contegration ranks. The presence of cointegration is a fundamental condition for the use of the VECM, as it confirms that the variables move together in the long run despite short-term fluctuations (Hall, 2011).

3.5.3 Optimal Lag Selection

Optimal lag selection is a fundamental step in time series modeling for all Vector Autoregressive models, including VECM. Selecting the appropriate number of lags balances the trade-off between model complexity and predictive accuracy. Choosing too few lags can lead to omitted variable bias, while too many lags may cause overfitting and a loss of degrees of freedom (Hall, 2011). In this study, the Akaike Information Criterion (AIC) is used to select the optimal lags.

3.6 Post-Estimation Diagnostics

To ensure the validity and reliability of the results for inferences, after estimating the VECM, the following tests will be performed:

3.6.1 Heteroscedasticity Test

Homoscedasticity assumes constant error variance across observations. Violation (heteroscedasticity) can bias standard errors, increasing the risk of Type I or Type II errors (Gujarati, 2012). The White test checks this assumption, where failure to reject the null indicates constant variance.

3.6.2 Multicollinearity Test

Multicollinearity occurs when some or all explanatory variables in a regression model are highly or perfectly linearly related (Gujarati, 2012). Perfect multicollinearity makes regression coefficients indeterminate with infinite standard errors, while high but imperfect multicollinearity results in large standard errors, reducing the precision of coefficient estimates. VECM assumes no collinearity among regressors, as it is estimated using differenced variables. This study tests for multicollinearity using pairwise correlation analysis; correlations above 0.7 indicate serious multicollinearity (Gujarati, 2012).

3.6.3 Autocorrelation/Serial Correlation Test

Serial correlation occurs when regression residuals are related to their own past values, which is a common problem in time-series data. When serial correlation is present, coefficient estimates remain unbiased but become inefficient, and the standard errors may be inflated. This issue is especially important in models that include lagged dependent variables (Enders, 2014). To test the null hypothesis that there is no serial correlation up to a chosen lag length the Breusch-Godfrey LM test will be used.

3.6.4 Normality Test

The Jarque-Bera (JB) test is used to assess whether regression residuals are normally distributed, based on their skewness and kurtosis. Normality is important for reliable statistical inference, especially in small samples, and violations may affect the accuracy of test statistics and confidence intervals (Chevillon, 2020).

3.6.5 Stability Test

The VECM's long-run stability is tested using the Hasza eigenvalue test, which checks whether all eigenvalues lie within the unit circle (Hansen & Johansen, 1999). If any eigenvalue is ≥ 1, the model is unstable; otherwise, it is stable. Stability is essential for reliable long- and short-run dynamics.

3.7 Economic and Statistical Packages Used for Analysis

All data analysis and econometric modeling will be performed using STATA 17. Microsoft-Excel 2019 will be used for data management.

3.8 Conclusion

This chapter outlines a robust methodology to examine the impact of agricultural supply chain disruptions on inflation in Malawi. Using a VECM framework and diagnostic tests, the study can analyze short- and long-run relationships, providing insights for policymakers to stabilize inflation and improve food security.

CHAPTER FOUR
RESULTS AND DISCUSSION

4.0 Introduction

This chapter presents the empirical results from the VECM specified in Chapter Three, highlighting the impact of agricultural supply chain disruptions on Malawi's headline inflation. Analyses were conducted using STATA 17. The chapter covers descriptive statistics, model diagnostics, and short- and long-run estimation results with their interpretations.

4.1 Descriptive Statistics

Table 2 presents the descriptive statistics for the variables used in the study. As noted by Groebner (2018), the mean represents the average value of each variable, while the standard deviation measures how far typical observations deviate from that mean (Anderson, 2011). The minimum and maximum values reflect the lowest and highest recorded values in the dataset, indicating the overall range (Anderson, 2011). A total of 55 observations were used for each variable over the 1970-2024 period.

Table 2: Descriptive Statistics

Variable	Obs.	Mean	Std. Dev.	Min	Max
Headline Inflation	55	17.223	13.243	2.821	83.326
Annual Rainfall	55	1074.759	155.208	729.35	1401.25
Fertilizer Price	55	192.045	133.321	16	700.02
Diesel Price	55	333.5	611.114	1.9	2734
Exchange Rate	55	208.145	362.296	.802	1751
Money Supply	55	4.260e+11	9.899e+11	47163000	5.329e+12

Source: Author's own computation using STATA 17 (Secondary data, 1970-2024).

Headline inflation rate averages of 17.2 percent, with a standard deviation of 13.2**,** indicating considerable volatility over the sample period. Inflation ranges from 2.8 percent to 83.3 percent, reflecting persistently high and unstable inflation which may hinder economic growth, undermine market confidence, worsen poverty, and weaken the domestic currency (Simwaka et al., 2012).

Annual rainfall averaged 1,074.8 mm, with a standard deviation of 155.2 mm, suggesting high variability in climatic conditions. It ranged from 729.4 mm to 1,401.3 mm. The mean suggests normal rainfall patterns, while the minimum signals drought conditions and the maximum reflects episodes of excessive rainfall and flooding (Mungai, 2016).

Fertilizer prices averaged USD 192.05 per metric ton, with a relatively high standard deviation of 133.321, and ranged from USD 16 to USD 700.02, indicating substantial price fluctuations. Diesel prices also shows high volatility, with a mean of MWK 333.5 and values ranging from MWK 1.9 to MWK 2,734, reflecting exposure to global oil price movements and domestic pricing policy shifts (Khonje, 2010).

The exchange rate averaged MWK 208.15 per USD, with a large standard deviation of 362.30, and fluctuated between MWK 0.80 and MWK 1,751, indicating significant exchange rate instability that may affect trade and inflation dynamics (Cleeve, 2016). Money supply recorded a mean value of MWK 426 billion, with a wide range from MWK 47 million to MWK 5.33 trillion, highlighting substantial monetary expansion over the study period (Simwaka et al., 2012).

4.2 Model diagnostic checks

To ensure that the model results are valid and reliable for inference and policy recommendations, the following diagnostic tests passed.

4.2.1 Optimal Lag Order Selection

Akaike Information Criterion (AIC) was employed as the primary criterion due to its effectiveness in balancing model fit with parsimony. The optimal lag length identified is four lags, corresponding to the minimum AIC value of 27.6533*, supported by a statistically significant likelihood ratio (LR) statistic of 90.001* with p-values of 0.000, indicating improved model fit with the inclusion of up to four lags. Refer to Appendix C for detailed lag test results.

4.2.2 Unit Root Test

Table 3: Unit Root Test using ADF

Variable	Lag	ADF Test	TCV (5%)	P-Value	Conclusion
			Level I(0)		
Headline Inflation	4	-2.546	-3.500	0.3052	Non-Stationery
Annual Rainfall	4	-2.968	-3.500	0.1411	Non-Stationery
Fertilizer Price	4	-2.305	-3.500	0.4313	Non-Stationery
Diesel price	4	-2.122	-3.500	0.5340	Non-Stationery
Exchange Rate	4	-2.670	-3.500	0.2489.	Non-Stationery
Money Supply	4	-3.146	-3.500	0.0958.	Non-Stationery
			Level I(1)		
Headline Inflation	0	-3.769	-2.928	0.0032	Stationery
Annual Rainfall	0	-7.796	-2.928	0.0000	Stationery
Fertilizer Price	0	-8.132	-2.928	0.0000	Stationery
Diesel price	0	-5.493	-2.928	0.0000	Stationery
Exchange Rate	0	-4.262	-2.928	0.0005	Stationery
Money Supply	0	-6.160	-2.928	0.0000	Stationery

Table 3 shows the results of the Augmented Dickey-Fuller (ADF) test applied to examine the stationarity of the variables, which is a prerequisite for estimating a VECM. The test was performed at both the levels (I(0)) and first differences (I(1)) with an optimal lag length of four for levels and zero for first differences.

At level, all variables; headline inflation, rainfall variability, fertilizer price, diesel price, exchange rate, and money supply, fail to reject the null hypothesis of a unit root. This is evidenced by ADF test statistics less negative than the 5% critical value (-3.50) and p-values above 0.05, indicating non-stationarity. Non-stationarity implies that these series exhibit time-dependent properties that undermine the assumptions needed for conventional time series analysis (Enders, 2014). However, after first differencing, all variables become stationary, with ADF statistics more negative than the 5% critical value (-2.928) and p-values below 0.05, confirming integration of order one, I(1).

4.2.3 Cointegration Test

Since the I(1) condition is satisfied, the study employed the Johansen cointegration test to confirm the existence of long-run equilibrium relationships and to identify the appropriate number of cointegrating equations for the model.

Table 4: Johansen Cointegration Test

Maximum Rank	Params	Trace Statistic	Critical-value 5%
0	114	154.2763	94.15
1	125	98.1885	68.52
2	134	58.9218	47.21
3	141	26.1538*	29.68
4	146	9.2162	15.41
5	149	0.0951	3.76
6	150		

The Johansen cointegration test evaluates the number of cointegrating vectors (ranks) based on the trace statistics compared to the 5% critical values. From Table 4, the trace statistics decline from 154.28 at rank 0 to 0.10 at rank 5. At rank 3, the trace statistic (26.15) falls below the critical value (29.68), indicating three cointegrating relationships among the variables. Identifying this rank is essential for specifying the VECM, as it captures both short-run dynamics and long-run equilibrium. The Johansen test provides a robust method for determining the number of cointegrating equations in multivariate time series (Enders, 2014).

4..2.4 Heteroskedasticity Test

The White test for heteroscedasticity yielded a chi-square of 16.38; p-value = 0.2290, indicating no evidence to reject the null of homoscedasticity at the5% level. This confirms constant error variance and supports the validity of the model’s standard errors (see Appendix E).

4.2.5 Multicollinearity Test

Following Gujarati's (2012) approach, pairwise correlation analysis was used to assess multicollinearity, since VECM does not support direct mean VIF estimation. The results show that all correlations are low, ranging from 0.0026 to 0.4519, well below the 0.7 threshold, indicating no serious multicollinearity, ensuring the validity and reliability of the estimates; (Appendix F).

4.2.6 Residual Serial Correlation /Auto correlation

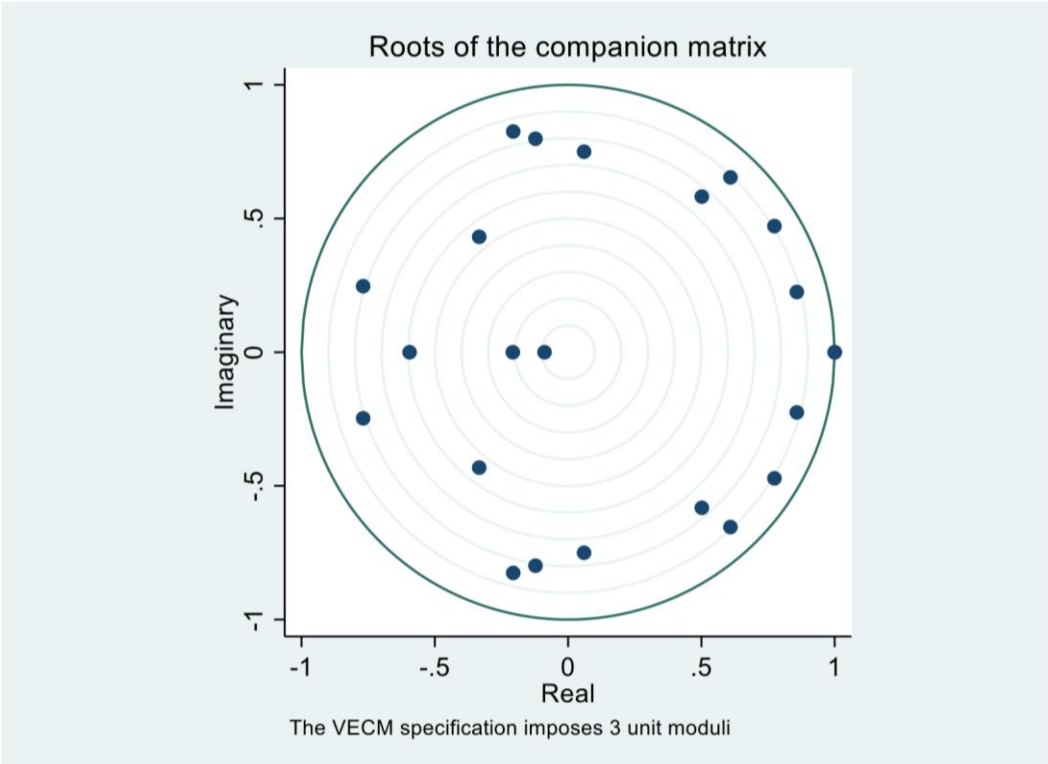
The Breusch-Godfrey LM test indicates no serial correlation in the VECM residuals across all lags. Chi-square statistics and p-values are $\chi^2 = 47.24$ ($p = 0.0996$) at lag 1, $\chi^2 = 43.57$ ($p = 0.1805$) at lag 2, $\chi^2 = 34.93$ ($p = 0.5195$) at lag 3, and $\chi^2 = 28.94$ ($p = 0.7920$) at lag 4. All p-values exceed 0.05, so the null of no serial correlation cannot be rejected, confirming that the VECM is well specified and parameter estimates are reliable (see Appendix G).

4.2.7 Residual Normality

The Jarque-Bera test for normality yielded $\chi^2 = 0.4331$ ($p = 0.8053$) statistically significant at the 5% level, indicating that the residuals are normally distributed. This confirms that the VECM estimates are unbiased and consistent (see Appendix H).

4.2.8 Stability Test

Figure 4: Companion Matrix



The Hasza criterion Companion Matrix in Figure 4 confirms all eigenvalues lie within the unit circle and all moduli ≤ 1, satisfying the stability condition and indicating that the model is dynamically stable, capturing both short- and long-run dynamics reliably.

4.3 VECM model results

VECM for the D_headline inflation equation shows strong explanatory power for short-run inflation dynamics in Malawi. With 51 observations and 22 parameters, the model records an RMSE of 7.001, indicating modest prediction error given inflation volatility. The R^2 of 0.9445 (adjusted $R^2 = 0.9011$) suggests that the regressors explains approximately 94.5% short-run inflation variation without overfitting. Overall significance is confirmed by the Wald chi-square statistic ($\chi^2 = 476.069$, $p < 0.0000$), indicating that the included variables jointly and significantly explain inflation movements.

4.3.1 Short-Run Dynamics

Table 5: Short-Run Dynamics (D_Headline Inflation)

Variable	Lag	Coefficient	Std. err.	Z	P>|z|
D_Headline Inflation	_ce1 L1.	-.4777184	.1773382	-2.69	0.007 ***
Headline Inflation	LD.	.1974176	.1941662	1.02	0.309
	L2D.	-.1955068	.2418745	-0.81	0.419
	L3D.	.4260747	.1494137	2.85	0.004 ***
Annual Rainfall	LD.	-.0098066	.0089362	-1.10	0.272
	L2D.	-.0261912	.0099951	-2.62	0.009 ***
	L3D.	-.0224475	.0085482	-2.63	0.009 ***
Diesel price	LD.	20.02848	8.634504	2.32	0.020 **
	L2D.	15.74023	8.524295	1.85	0.065 *
	L3D.	25.07993	9.123838	2.75	0.006 ***
Fertilizer Price	LD.	.0139151	.0351216	0.40	0.692
	L2D.	.0269292	.0305258	0.88	0.378
	L3D.	.0182134	.0275589	0.66	0.509
Exchange Rate	LD.	51.01785	13.81998	3.69	0.000 ***
	L2D.	8.119243	16.61692	0.49	0.625
	L3D.	24.5242	15.30617	1.60	0.109
Money Supply	LD.	10.17905	12.05711	0.84	0.399
	L2D.	-15.75767	12.00777	-1.31	0.189
	L3D.	-29.13365	12.32911	-2.36	0.018 **
_cons		.1037952	61.19425	0.00	0.999

**** p<.01, ** p<.05, * p<.1*

The Error Correction terms (_ce)

The error-correction coefficient (_ce1, L1 = -0.4777, p = 0.007) is negative and statistically significant at the 1% level, confirming a stable long-run equilibrium. Approximately 48% of deviations from long-run inflation are corrected within one period, indicating rapid adjustment.

This shows that Malawi's inflation responds strongly to shocks from climatic conditions, fuel and fertilizer prices, exchange rates, and monetary factors, and is endogenously driven back to its long run path, consistent with VECM theory (Johansen, 1995).

Lagged Headline Inflation (Inertial)

The lagged inflation terms indicate limited short-run persistence but significant delayed inertia. While the first and second lags are insignificant, the third lag is positive and significant at the 1% level (L3D = 0.4261, p = 0.004), implying that a one-percentage-point increase in inflation three periods earlier raises current inflation by about 0.43 percentage points. This delayed persistence suggests gradual transmission of inflationary shocks, reflecting price rigidities and backward-looking expectations common in developing economies characterized by structural rigidities and imperfect markets (Bresser-Pereira & Nakano, 2003), and is consistent with evidence from Pakistan (Hanif, 2016).

Annual Rainfall

The first lag is statistically insignificant at all three conventional levels of significant (LD = 0.0098, p = 0.272), while the second and third lags are both negative and significant at the 1% level (L2D = -0.0262, p = 0.009; L3D = -0.0224, p = 0.009). Quantitatively, these coefficients imply that a one-unit improvement in rainfall conditions reduces headline inflation by about 0.03 percentage points after two periods and by about 0.02 percentage points after three periods, holding other factors constant. This delayed effect reflects agricultural production cycles: adverse rainfall conditions reduce crop yields, tighten food supplies, and raise prices after harvest periods, whereas favorable rainfall improves output, eases supply constraints, and dampens inflation.

These findings align with recent empirical evidence from other agriculture-dependent economies. For example, Cevik and Jalles (2023) show that rainfall variability and temperature shocks significantly increase inflation while putting downward pressure on the exchange rate in developing countries.

Fertilizer Price

The results (p-values = 0.692, 0.378, 0.509 respectively) indicate that fertilizer prices do not have a meaningful impact on headline inflation in Malawi, as all the coefficients are statistically insignificant across all lags at all conventional levels of statistical significance. Although the study anticipated a positive relationship based on standard cost-push inflation theory that rising fertilizer prices increase production costs and eventually consumer prices, the study's empirical evidence indicates that this theoretical expectation does not hold in Malawi's structural environment.

This pattern aligns with findings from countries with extensive fertilizer subsidy programmes, where the pass-through from international fertilizer prices to domestic consumer prices is weakened. Malawi's long-standing subsidy initiatives, including Starter Pack, AIP, and FISP, have historically insulated smallholder farmers from global fertilizer price volatility. As shown by Chibwana (2016), the effective fertilizer cost for most farmers is set by government-subsidised prices rather than market prices, thereby limiting fertilizer’s inflationary impact. Comparable evidence from Zambia (Chapoto et al., 2015), Tanzania (Minot, 2009), and Nigeria (Olayemi, 2012) similarly shows that subsidies disrupt the transmission of global input costs to domestic inflation.

Remittances within Malawi's extended family culture and high dependency ratio also play a role, as Malawians from abroad and urban areas send money to relatives during farming seasons to purchase inputs, especially expensive fertilizers, shielding them from actual market prices. However, beneficiaries likely do not take market costs into account since they did not bear the burden themselves and may still sell output at lower prices. Studies such as "Migrant Remittances and Household Food Security" by Riley et al. (2022) and Chami (2008) provide evidence that remittances act as a shock-absorbing mechanism in developing economies, stabilizing household consumption and smoothing income during adverse macroeconomic events, including high input prices. This is also consistent with the World Bank (2022) country report, which notes that Malawi's inflation is mainly driven by exchange-rate depreciation, imported fuel costs, and maize supply shocks rather than fertilizer prices. Therefore, although a positive effect was theoretically expected, the insignificant coefficients in the VECM model are empirically plausible and consistent with regional evidence.

Diesel Prices

Diesel price dynamics exert strong and statistically significant effects on headline inflation across all lags, highlighting the importance of energy costs in Malawi's price formation. The first lag (LD = 20.03, p = 0.020) is positive and significant at the 5% level: a one Malawi kwacha increase in diesel pump prices in the previous year increases current headline inflation by about 20.03 percentage points, indicating immediate pass-through higher transport, production, and distribution costs. The second lag (L2D = 15.74, p = 0.065) is positive and marginally significant at the 10% level: a one-kwacha increase in diesel pump prices two periods ago is associated with a 15.74 percentage point increase in headline inflation, suggesting continued adjustment as businesses revise prices in line with accumulated cost pressures. The third lag (L3D = 25.08, p = 0.006) is positive and significant at the 1% level: a one-kwacha increase in diesel prices three periods ago leads to a 25.08 percentage point increase in current headline inflation, confirming a strong and persistent multi-period pass-through effect. This pattern indicates that diesel price shocks propagate gradually through the economy, consistent with cost-push inflation and staggered price-setting behaviour. These findings align with Babuga (2022), who shows that in sub-Saharan Africa, oil price increases have a stronger impact on inflation than decreases, highlighting an asymmetric effect. Overall, the results underscore the central role of fuel costs in driving Malawi's short- to medium-term inflation dynamics.

Exchange Rate

Exchange rate coefficients show strong and immediate short-run effects on headline inflation, reflecting Malawi’s heavy reliance on imported goods and foreign-currency-priced inputs. The first lag is large, positive, and highly significant (LD = 51.02, p = 0.000), implying that a one-percent depreciation of the Malawi kwacha against the US dollar raises headline inflation by about 51 percentage points, reflecting rapid pass-through to domestic prices, especially in import dependent sectors such as fuel, raw materials, and manufactured goods. In contrast, the second and third lags are positive but statistically insignificant at all conventional levels, indicating limited adjustment beyond the initial shock. Overall, exchange rate pass-through in Malawi is largely front-loaded, consistent with shallow foreign exchange markets, high import dependence, and limited pricing-to-market behaviour, in line with evidence from sub-Saharan Africa (IMF, 2024; Harvey & Cushing, 2013).

Money Supply

The first and second lags of money supply are statistically insignificant at all conventional levels, indicating weak immediate effects on headline inflation. However, the third lag is negative and statistically significant at the 5% level (L3D = -29.13, $p = 0.018$), implying that a one-unit increase in broad money supply three periods earlier is associated with an approximate 29 percentage-point reduction in current headline inflation. This suggests a weak and delayed monetary transmission mechanism in the short run, where monetary changes do not translate immediately into price pressures and unpredictable money velocity.

In theory, money supply is inversely related to interest rates: an expansion in money supply lowers interest rates, stimulates borrowing for investment and consumption, and boosts aggregate demand and output in the short run. If output responds sufficiently, prices may remain relatively stable initially; however, sustained monetary expansion eventually generates inflationary pressures in the long run as aggregate demand outpaces aggregate supply, consistent with the *"too much money chasing too few goods"* argument (Mishkin, 2012). The observed pattern indicates that money supply effects on inflation in Malawi operate mainly over a medium-term horizon, reflecting slow demand adjustment, structural rigidities, and weak transmission channels.

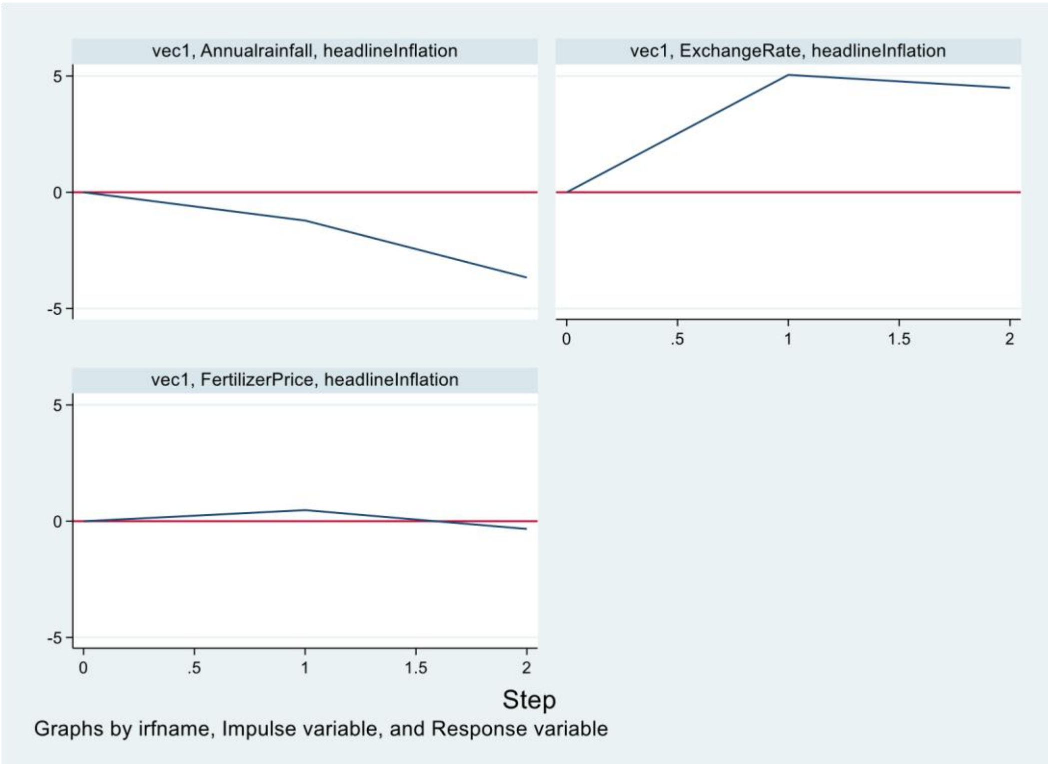
These findings are consistent with evidence from Ghana and Nigeria, where Adusei (2013) and Gatawa (2017) show that broad money has weak or even negative short-run effects on inflation but a positive long-run impact, indicating that monetary expansion initially stimulates real activity before price effects dominate. Similar results are reported for sub-Saharan Africa, where money supply changes influence inflation with lags due to adaptive expectations, liquidity constraints, and weak monetary transmission channels (Ndung'u & Ngugi, 2007; Nachega, 2005). Overall, the results imply that the immediate inflationary effects of money supply dynamics in Malawi are limited.

4.3.2 Impulse Response Functions

The impulse response functions (IRFs) show the short-run or immediate effects of shocks in the VECM system. They trace how headline inflation responds to a one-unit or one standard deviation shock in each variable, while keeping other shocks constant.

Figure 5 shows how headline inflation responds to shocks from exchange rate fluctuations, fertilizer prices, weather and climatic conditions proxied by annual rainfall.

Figure 5: Impulse Response Functions



The IRFs in Figure 5 confirms the model's findings. A positive annual rainfall shock (better rains) reduces headline inflation, as the response remains negative throughout the two-period horizon, indicating a persistent negative effect. An exchange rate shock causes a sharp rise in inflation in the first period, then stabilizes in the second period while remaining above zero, confirming a positive relationship whereby depreciation of the Malawi kwacha increases headline inflation. Fertilizer price shocks have no meaningful effect on headline inflation, as the response remains fairy around zero over the entire two-year horizon. Overall, the IRFs reinforce the VECM results by highlighting the dominant inflationary impact of exchange rate movements and disinflationary impact of climatic conditions, and insignificance of fertilizer prices.

4.3.3 Long Run Analysis

Table 6: Long-Run Dynamics **(**Johansen normalization restrictions imposed)

Beta	**Coefficient**	**Std. err.**	**z**	**P>|z|**
_ce1	1			
Headline Inflation		.	.	.
Annual Rainfall	-6.51e-18	.	.	.
Fertilizer Price	6.94e-17	.	.	.
Diesel price	-42.03469	19.66005	-2.14	0.033 **
Exchange Rate	-232.2575	29.95841	-7.75	0.000 ***
Money Supply	107.1309	30.97331	3.46	0.001 ***
_cons	-2236.266	.	.	.
_ce2				
Headline Inflation	2.13e-14	.	.	.
Annual Rainfall	1	.	.	.
Fertilizer Price	7.11e-15	.	.	.
Diesel price	-6192.726	1271.808	-4.87	0.000 ***
Exchange Rate	-11208.03	1938.008	-5.78	0.000 ***
Money Supply	6838.01	2003.662	3.41	0.001 ***
_cons	-128810.2	.	.	.
_ce3				
Headline Inflation	2.22e-1	.	.	.
Annual Rainfall	-3.47e-18	.	.	.
Fertilizer Price	1	.	.	.
Diesel price	97.65122	19.32525	5.05	0.000 ***
Exchange Rate	302.2286	29.44823	10.26	0.000 ***
Money Supply	-295.5811	30.44585	-9.71	0.000 ***
_cons	6134.907	.	.	.

*** $p<.01$, ** $p<.05$, * $p<.1$

Interpretation of Long-Run Coefficients in a VECM

In a VECM framework, long-run coefficients are interpreted with reversed signs due to normalization of the cointegrating vectors (Rehal, 2022).

A cointegrating relationship is generally expressed as:

$$\beta' \mathbf{X}_t = 0$$

If this relationship holds, then multiplying the entire equation by -1 yields:

$$-\beta' \mathbf{X}_t = 0$$

which is equally valid. This sign reversal is normal and purely mechanical rather than econometric error and reflects the fact that cointegrating relationships are identified only up to a scalar multiple (Hapsari, 2021). Consequently, reversing the signs allows the long-run equilibrium relationship to be expressed conventionally, with headline inflation written as a function of its long-run determinants.

Due to Johansen normalization, STATA does not report standard errors or p-values for lagged headline inflation, annual rainfall, and fertilizer prices; therefore, their long-run effects cannot be statistically interpreted. In contrast, diesel prices, the exchange rate, and money supply display statistically significant long-run coefficients.

From **Table 6:** Extracted from STATA output**,** the three estimated cointegrating equations are:

Inflation −42.034 Diesel − 232.258Exchange Rate +107.131Money Supply = 0…(_ce1)

Inflation −6192.73 Diesel − 11208.03Exchange Rate +6838.01Money Supply = 0..(_ce2)

Inflation +97.651 Diesel +302.228Exchange Rate − 295.581Money Supply = 0…(_ce3)

Reversing these equation yields the normalized long-run relationships as;

Inflation = 42.034 Diesel +232.258Exchange Rate −107.131Money Supply…(_ce1)

Inflation = 6192.73 Diesel + 11208.03Exchange Rate −6838.01Money Supply…(_ce2)

Inflation= −97.651 Diesel −302.228Exchange Rate + 295.581Money Supply …(_ce3)

Cumulative effect of the normalized long-run relationships;

Inflation= 6137.109Diesel + 11138.059Exchange Rate + 295.581Money Supply

Interpretation follows the principle that larger coefficients in the normalized cumulative cointegrating equations reflect stronger long-run impacts (Hapsari, 2021; Rehal, 2022). The results indicate that diesel prices, the exchange rate, and money supply are statistically significant at the 1% level ($p < 0.01$), confirming their relevance in explaining long-run headline inflation dynamics in Malawi. Among these variables, exchange rate movements emerge as the dominant long-run driver of inflation, followed by monetary expansion and fuel price increases.

Specifically, the first cointegrating equation shows that a percentage increase in diesel prices is associated with approximately a 42 percentage increase in headline inflation in the long run, while a percentage depreciation of the Malawian Kwacha leads to nearly a 232 percent increase in inflation. This suggests a high vulnerability of inflation to transport and production costs, as well as increased landed costs of key imported inputs, which ultimately translate into higher consumer prices. The second equation further reinforces the dominance of exchange rate depreciation and diesel price increases in intensifying long-run inflationary pressures. In contrast, the third equation indicates that monetary expansion exerts a substantial inflationary effect, with a percentage increase in money supply raising headline inflation by about 296 percent in the long run.

CHAPTER FIVE

CONCLUSIONS AND POLICY RECOMMENDATIONS

5.1 Introduction

This chapter summarizes the main findings on the impact of agricultural supply chain disruptions on headline inflation in Malawi. It further presents policy recommendations to address structural vulnerabilities, discusses key limitations of the study, and proposes areas for future research to deepen understanding of inflation dynamics in agriculture-dependent economies.

5.2 Conclusion

The study set out to investigate the impact of agricultural supply chain disruptions on headline inflation in Malawi using annual time series data spanning 1970-2024. Specifically, it tested the null hypotheses that climatic shocks (proxied by annual rainfall), fertilizer price dynamics, and that fuel price changes do not significantly affect headline inflation. The analysis employed a VECM framework to capture both short-run dynamics and long-run equilibrium relationships.

The empirical results provide strong evidence that, at the 5 percent significance level, climatic shocks and fuel price changes exert a statistically significant influence on headline inflation in both the short and long run. Accordingly, the null hypotheses that annual rainfall has no relationship with headline inflation and that fuel prices have no effect on inflation are both rejected. In contrast, the study fails to reject the null hypothesis that fertilizer price dynamics do not significantly contribute to headline inflation, suggesting that fertilizer prices have no statistically significant direct impact on inflation, likely reflecting the cushioning effects of remittances and government subsidies.

Overall, the findings indicate that headline inflation in Malawi is predominantly supply-driven, with agricultural supply chain disruptions particularly climatic and weather-related shocks, and fuel price volatility playing a central role in shaping inflation dynamics. Exchange rate movements further amplify these effects by transmitting external price pressures into the domestic economy. Reinforcing the conclusion that controlling inflation requires more than monetary tightening and praying for good rains; it also requires strengthening agricultural supply chains, by improving energy logistics, and building resilience to climate shocks.

5. 3 Policy Recommendations and Implications

The study findings underscore the need for a multi-sectoral approach to controlling inflation in Malawi, driven by supply chain disruptions and structural inefficiencies. Policy responses should therefore go beyond conventional monetary tightening and focus on strengthening the agricultural and energy sectors, as well as exchange rate management. These recommendations align with Pillar One of Malawi Vision 2063 on Agricultural Productivity and Commercialization, the Sustainable Development Goals particularly SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action) and the Reserve Bank of Malawi's medium-term headline inflation target of 5 percent.

Strengthening climate resilience in agriculture is critical. Given the significant impact of rainfall variability on inflation, the government should invest in irrigation infrastructure, climate-smart agriculture, and early warning systems to reduce dependence on rain-fed production. This includes Intensifying modern irrigation, promoting drought-resistant crop varieties, and expanding mechanization to reduce time losses and improve production efficiency, which would stabilize food supply, and reduce inflationary pressures arising from climatic shocks.

Policies to curb fuel cost volatility are essential, as fuel prices directly impact headline inflation through production and transport costs. Expanding and modernizing fuel reserves, improving procurement and logistics, as well as promoting renewable energy sources like ethanol and biodiesel can lower reliance on imported fuel and reduce supply chain costs. Upgrading rural roads would further reduce transport costs, easing diesel-driven inflationary pressures through agricultural distribution.

Thirdly, maintaining exchange rate stability is essential due to its strong long-run effect on inflation. Prudent foreign exchange management, export diversification, reserve accumulation, and import hedging for critical inputs, alongside promoting value addition and export-oriented agro-processing, can reduce external vulnerabilities and strengthen macroeconomic stability.

Finally, while fertilizer prices were found to be statistically insignificant, maintaining welltargeted and fiscally sustainable input subsidy programs remains important for supporting agricultural productivity and food security. However, such programs should be complemented with broader supply chain investments to ensure long-term price stability.

5.4 Limitations of the Study

Despite its contributions, this study has several limitations. First, the analysis relies on annual data, which may miss short-term dynamics and adjustment processes that could be captured using higher-frequency data, such as quarterly or monthly observations. Second, rainfall used as a proxy for climatic shocks, may not fully capture other relevant climate-related factors, including temperature extremes and the frequency of extreme weather events. Finally, data constraints limited the inclusion of additional relevant variables, such as drought duration, and black market fuel prices, which may also influence inflation dynamics.

5.5 Areas for Future Research

Future research could extend this study in several directions. Using higher-frequency data would allow for a more detailed examination of short-run inflation dynamics and the timing of supply chain shocks. Incorporating additional climate indicators, such as temperature variability or drought indices, would provide a more comprehensive assessment of climate-related inflationary pressures. If more data become available, further studies could also explore non-linear models or structural break tests to assess whether the impact of agricultural supply chain disruptions on inflation varies across different policy regimes or economic conditions.

REFERENCES

Adeniyi, O., & Yusuf, A. (2023). Oil price shocks and food inflation in Nigeria: A nonlinear ARDL approach. *International Journal of Energy Economics and Policy, 13*(2), 128-134.

Adusei, M. (2013). Financial development and economic growth: Evidence from Ghana. *The International Journal of Business and Finance Research, 7*(5), 61-76.

Adusei, M. (2013). Is inflation in South Africa a structural or monetary phenomenon? *Journal of Economics, Management and Trade*, *3*(1), 60-72. https://doi.org/10.9734/BJEMT/2013/2553

AfricaFertilizer.org, Development Gateway, & International Fertilizer Development Center. (2022). *Malawi fertilizer statistics overview 2022 edition*.

Anderson, D. R., Sweeney, D. J., & Williams, T. A. (2011). *Statistics for business and economics* (11th ed.). South-Western Cengage Learning.

Asteriou, D., & Hall, S. G. (2011). *Applied econometrics* (2nd ed.). Palgrave Macmillan.

Babuga, U. T., & Naseem, N. A. M. (2022). Asymmetric effect of oil price change on inflation: Evidence from sub-Saharan Africa countries. *International Journal of Energy Economics and Policy*, *11*(1), 448-458. https://doi.org/10.32479/ijeep.10764

Belbute, J. M., Massala, L. D., & Delgado, J. A. (2015). Measuring persistence in inflation: Evidence from Angola. *Journal of African Economies*, *24*(S1), i6-i95.

Bresser-Pereira, L. C., & Nakano, Y. (2003). *The theory of inertial inflation: The foundation of economic reform in Brazil and Argentina*. Lynne Rienner Publishers.

Cevik, S., & Jalles, J. T. (2023). Eye of the storm: The impact of climate shocks on inflation and growth. *International Monetary Fund Working Paper*.

Chapoto, A., Mason, N. M., & Jayne, T. S. (2015). How do input subsidy programs affect maize markets? Evidence from Zambia. *Food Policy*, *54*, 41-51.

Chavula, H. K. (2016). Monetary policy effects and output growth in Malawi: Using a small macroeconometric model. *International Journal of Economics and Finance, 8*(10), 98-110.

Chevillon, G., Mavroeidis, S., & Zhan, Z. (2020). Robust inference in structural vector autoregressions with long-run restrictions. *Econometric Theory, 36*(1), 86-121. https://doi.org/10.1017/S0266466619000045

Chirwa, E. W., & Chinsinga, B. (2011). *Maize price instability in Malawi: Implications for food security*. Future Agricultures Consortium Working Paper.

Chirwa, E. W., & Zakeyo, C. (2003). *Impact of agricultural input subsidies in Malawi*. Ministry of Agriculture & IFPRI.

Chopra, S., & Meindl, P. (2016). *Supply chain management: Strategy, planning, and operation* (6th ed.). Pearson.

Dawe, D. (2008). Have recent increases in international cereal prices been transmitted to domestic economies? *FAO ESA Working Paper No. 08-03*.

Dolgui, A., Ivanov, D., & Sokolov, B. (2018). Ripple effect in the supply chain: An analysis and recent literature. *International Journal of Production Research, 56*(1-2), 414-430.

Enders, W. (2014). *Applied econometric time series* (4th ed.). John Wiley & Sons.

Food and Agriculture Organization of the United Nations. (2020). *The state of food security and nutrition in the world 2020*. FAO.

Friedman, M., & Schwartz, A. J. (1963). *A monetary history of the United States, 1867-1960*. Princeton University Press.

Galí, J. (2008). *Monetary policy, inflation, and the business cycle: An introduction to the New Keynesian framework*. Princeton University Press.

Gillman, M. (2009). *Inflation theory in economics: Welfare, velocity, growth and business cycles*. Routledge.

Gordon, R. J. (2012). *Macroeconomics* (12th ed.). Pearson.

Gujarati, D. N. (2012). *Basic econometrics* (5th ed.). McGraw-Hill Education.

Hanif, M. N. (2016). Intrinsic inflation persistence in a developing country. *SBP Research Bulletin, 12*(2).

Hapsari, M. R., Astutik, S., & Soehono, L. A. (2021). VECM and Bayesian VECM for overparameterization problem. *Journal of Physics: Conference Series, 1811*(1), 012086. https://doi.org/10.1088/1742-6596/1811/1/012086

Hobbs, J. E. (2020). Food supply chains during the COVID-19 pandemic. *Canadian Journal of Agricultural Economics, 68*(2), 171–176.

International Monetary Fund. (2024). *Effect of exchange rate movements on inflation in SubSaharan Africa* (Working Paper No. WP/24/059). https://www.imf.org//media/files/publications/wp/2024/english/wpiea2024059-print-pdf.pdf

Jayne, T. S., Ricker-Gilbert, J., Mason, N. M., & Chibwana, C. (2016). Subsidizing agriculture: Input subsidies in Africa. *Applied Economic Perspectives and Policy, 38*(1), 113-138.

Jenkins, S. (2023). The Russia–Ukraine war's impact on fertilizer prices in 2022. *On the Economy Blog*. Federal Reserve Bank of St. Louis.

Jones, R. (2018). Hyperinflation in history: Weimar Germany and Zimbabwe. *Economic History Review*.

Kalua, B. (2017). Agricultural determinants of inflation in Malawi. *Journal of Economics and Sustainable Development, 8*(12), 45-56.

Khonje, M. G. U. (2010). Food inflation in Malawi: Implications for the economy (Master's thesis, Makerere University).

Khonje, M. G. (2020). Determinants of inflation in Malawi. *African Journal of Economic and Management Studies, 11*(3), 345-362.

Kunawotor, M. E., et al. (2022). Climatic shocks and agricultural yields in sub-Saharan Africa. *Climate and Development*, *14*(5), 456-468.

Mangani, R. (2013). External factors and inflation in Malawi. *Reserve Bank of Malawi Working Paper*.

Mankiw, N. G. (2021). *Principles of macroeconomics* (9th ed.). Cengage Learning.

McEachern, W. A. (2006). *Economics: A contemporary introduction* (7th ed.). Thomson SouthWestern.

McKinsey & Company. (2024). *Inflation and inequality: Global trends and implications*. McKinsey Global Institute.

Minot, N. (2009). Transmission of world food price changes to markets in low-income countries. *IFPRI Discussion Paper*.

Mishkin, F. S. (2012). *Macroeconomics: Policy and practice* (2nd ed.). Pearson.

Mungai, L. M., Snapp, S., Messina, J. P., Chikowo, R., Smith, A., Anders, E., Richardson, R. B., & Li, G. (2016). Smallholder farms and the potential for sustainable intensification. *Frontiers in Plant Science, 7*, 1720. https://doi.org/10.3389/fpls.2016.01720

Nkomboni, D., & Ayal, D. (2022). Climate variability and price instability in rain-fed agriculture economies: Evidence from Malawi. *Climate Risk Management, 35*, 100412.

Our Reporter. (2024, February 19). *Inflation rises to 35% as pressures remain elevated*. Nation Online. https://mwnation.com/inflation-rises-to-35-as-pressures-remain-elevated

Paul, S. K., Agarwal, R., Sarker, R. A., & Rahman, T. (2023). *Supply chain risk and disruption management: Latest tools, techniques and management approaches* (1st ed.). Springer Nature Singapore.

Posen, A. S., Bernanke, B. S., Mishkin, F. S., & Laubach, T. (1999). *Inflation targeting: Lessons from the international experience*. Princeton University Press.

Rehal, V. (2022, May 9). *VECM estimation and interpretation*. SPUR Economics. https://spureconomics.com/vecm-estimation-and-interpretation/

Reserve Bank of Malawi. (2023). *Financial and economic review*. RBM.

Reserve Bank of Malawi. (2024). *Annual economic report*. RBM.

Self Help Africa. (2024). *El Niño drought impacts in Malawi: Situation report*.

Simwaka, K. (2018). Money supply and inflation in Malawi: An econometric investigation. *International Journal of Applied Economics and Finance, 6*(2), 74-88.

Tankari, M. R., & Fofana, I. (2024). Climatic shocks and food inflation in sub-Saharan Africa. *African Development Review, 36*(1), 123-140.

U.S. Department of Agriculture, Economic Research Service. (2022). *World agricultural output growth continues to slow*. USDA ERS.

Wauk, G., & Adjorlolo, G. (2019). The game of monetary policy, inflation and economic growth. *Open Journal of Social Sciences, 7*(3), 255-271. https://doi.org/10.4236/jss.2019.73022

World Bank. (2016). *Fuel price impacts on inflation in Malawi*. World Bank.

World Bank. (2023). *Global food inflation trends*. Commodity Markets Outlook.

World Inequality Lab. (2022). *World inequality report 2022*. Harvard University Press.

APPENDICES

APPENDIX A: VECM Estimation

```
Vector error-correction model

Sample: 1974 thru 2024                          Number of obs     =         51
                                                AIC               =   27.81319
Log likelihood = -568.2362                      HQIC              =   29.85411
Det(Sigma_ml)  =  191.8113                      SBIC              =   33.15412
```

Equation	Parms	RMSE	R-sq	chi2	P>chi2
D_headlineInfl~n	22	7.00128	0.9445	476.069	0.0000
D_Annualrainfall	22	142.742	0.9902	2833.37	0.0000
D_FertilizerPr~e	22	82.3538	0.5851	39.48071	0.0124
D_dieselprice	22	.147246	0.7711	94.34588	0.0000
D_ExchangeRate	22	.120463	0.8400	147.0189	0.0000
D_MoneySupply	22	.113113	0.8825	210.2677	0.0000

	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
D_headlineInflation						
_ce1						
L1.	-.4777184	.1773382	-2.69	0.007	-.8252948	-.130142
_ce2						
L1.	.0094346	.0027542	3.43	0.001	.0040365	.0148328
_ce3						
L1.	-.0375489	.0441205	-0.85	0.395	-.1240236	.0489258
headlineInflation						
LD.	.1974176	.1941662	1.02	0.309	-.1831411	.5779763
L2D.	-.1955068	.2418745	-0.81	0.419	-.6695721	.2785585
L3D.	.4260747	.1494137	2.85	0.004	.1332292	.7189201
Annualrainfall						
LD.	-.0098066	.0089362	-1.10	0.272	-.0273213	.007708
L2D.	-.0261912	.0099951	-2.62	0.009	-.0457812	-.0066011
L3D.	-.0224475	.0085482	-2.63	0.009	-.0392016	-.0056935
FertilizerPrice						
LD.	.0139151	.0351216	0.40	0.692	-.054922	.0827523
L2D.	.0269292	.0305258	0.88	0.378	-.0329003	.0867587
L3D.	.0182134	.0275589	0.66	0.509	-.0358011	.0722279
dieselprice						
LD.	20.02848	8.634504	2.32	0.020	3.10516	36.95179
L2D.	15.74023	8.524295	1.85	0.065	-.9670818	32.44754
L3D.	25.07993	9.123838	2.75	0.006	7.197539	42.96233
ExchangeRate						
LD.	51.01785	13.81998	3.69	0.000	23.9312	78.10451
L2D.	8.119243	16.61692	0.49	0.625	-24.44931	40.6878
L3D.	24.5242	15.30617	1.60	0.109	-5.475352	54.52374
MoneySupply						
LD.	10.17905	12.05711	0.84	0.399	-13.45244	33.81055
L2D.	-15.75767	12.00777	-1.31	0.189	-39.29246	7.777124
L3D.	-29.13365	12.32911	-2.36	0.018	-53.29827	-4.969036
_cons	.1037952	61.19425	0.00	0.999	-119.8347	120.0423

Cointegrating equations

Equation	Parms	chi2	P>chi2
_ce1	3	3324.12	0.0000
_ce2	3	2372.759	0.0000
_ce3	3	382.616	0.0000

Identification: beta is exactly identified

Johansen normalization restrictions imposed

beta	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]	
_ce1						
headlineInflation	1	.	.	.	.	.
Annualrainfall	-6.51e-18	.	.	.	.	.
FertilizerPrice	6.94e-17	.	.	.	.	.
dieselprice	-42.03469	19.66005	-2.14	0.033	-80.56767	-3.501701
ExchangeRate	-232.2575	29.95841	-7.75	0.000	-290.9749	-173.5401
MoneySupply	107.1309	30.97331	3.46	0.001	46.42428	167.8374
_cons	-2236.266	.	.	.	.	.
_ce2						
headlineInflation	2.13e-14	.	.	.	.	.
Annualrainfall	1	.	.	.	.	.
FertilizerPrice	7.11e-15	.	.	.	.	.
dieselprice	-6192.726	1271.808	-4.87	0.000	-8685.423	-3700.029
ExchangeRate	-11208.03	1938.008	-5.78	0.000	-15006.46	-7409.605
MoneySupply	6838.01	2003.662	3.41	0.001	2910.904	10765.12
_cons	-128810.2	.	.	.	.	.
_ce3						
headlineInflation	2.22e-16	.	.	.	.	.
Annualrainfall	-3.47e-18	.	.	.	.	.
FertilizerPrice	1	.	.	.	.	.
dieselprice	97.65122	19.32525	5.05	0.000	59.77443	135.528
ExchangeRate	302.2286	29.44823	10.26	0.000	244.5111	359.9461
MoneySupply	-295.5811	30.44585	-9.71	0.000	-355.2539	-235.9083
_cons	6134.907	.	.	.	.	.

APPENDIX B: ADF test for Unit Root (Stationarity)

```
. dfuller headlineInflation , lags(4) trend

Augmented Dickey-Fuller test for unit root

Variable: headlineInflat~n                 Number of obs  = 50
                                           Number of lags =  4

H0: Random walk with or without drift

                                       Dickey-Fuller
                   Test         ———————— critical value ————————
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -2.546       -4.150       -3.500       -3.180
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.3052.

.
. dfuller Annualrainfall , lags(4) trend

Augmented Dickey-Fuller test for unit root

Variable: Annualrainfall                   Number of obs  = 50
                                           Number of lags =  4

H0: Random walk with or without drift

                                       Dickey-Fuller
                   Test         ———————— critical value ————————
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -2.968       -4.150       -3.500       -3.180
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.1411.

.
. dfuller FertilizerPrice,lags(4) trend

Augmented Dickey-Fuller test for unit root

Variable: FertilizerPrice                  Number of obs  = 50
                                           Number of lags =  4

H0: Random walk with or without drift

                                       Dickey-Fuller
                   Test         ———————— critical value ————————
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -2.305       -4.150       -3.500       -3.180
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.4313.

.
. dfuller dieselprice , lags(4) trend

Augmented Dickey-Fuller test for unit root

Variable: dieselprice                      Number of obs  = 50
                                           Number of lags =  4

H0: Random walk with or without drift

                                       Dickey-Fuller
                   Test         ———————— critical value ————————
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -2.122       -4.150       -3.500       -3.180
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.5340.
```

```
. dfuller ExchangeRate , lags(4) trend

Augmented Dickey–Fuller test for unit root

Variable: ExchangeRate                    Number of obs  = 50
                                          Number of lags =  4

H0: Random walk with or without drift

                                     Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -2.670       -4.150       -3.500       -3.180
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.2489.

.
. dfuller MoneySupply , lags(4) trend

Augmented Dickey–Fuller test for unit root

Variable: MoneySupply                     Number of obs  = 50
                                          Number of lags =  4

H0: Random walk with or without drift

                                     Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -3.146       -4.150       -3.500       -3.180
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0958.

. dfuller D.headlineInflation

Dickey–Fuller test for unit root          Number of obs  = 53
Variable: D.headlineInfl~n                Number of lags =  0

H0: Random walk without drift, d = 0

                                     Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -3.769       -3.576       -2.928       -2.599
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0032.

.
. dfuller D.Annualrainfall

Dickey–Fuller test for unit root          Number of obs  = 53
Variable: D.Annualrainfall                Number of lags =  0

H0: Random walk without drift, d = 0

                                     Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -7.796       -3.576       -2.928       -2.599
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000.

.
```

```
. dfuller D.FertilizerPrice

Dickey–Fuller test for unit root          Number of obs  = 53
Variable: D.FertilizerPr~e                Number of lags =  0

H0: Random walk without drift, d = 0

                                       Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -8.132       -3.576       -2.928       -2.599
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000.

.
. dfuller D.dieselprice

Dickey–Fuller test for unit root          Number of obs  = 53
Variable: D.dieselprice                   Number of lags =  0

H0: Random walk without drift, d = 0

                                       Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -5.493       -3.576       -2.928       -2.599
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000.

.
. dfuller D.ExchangeRate

Dickey–Fuller test for unit root          Number of obs  = 53
Variable: D.ExchangeRate                  Number of lags =  0

H0: Random walk without drift, d = 0

                                       Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -4.262       -3.576       -2.928       -2.599
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0005.

.
. dfuller D.MoneySupply

Dickey–Fuller test for unit root          Number of obs  = 53
Variable: D.MoneySupply                   Number of lags =  0

H0: Random walk without drift, d = 0

                                       Dickey–Fuller
                   Test      ──────── critical value ─────────
              statistic           1%           5%          10%
──────────────────────────────────────────────────────────────
 Z(t)            -6.160       -3.576       -2.928       -2.599
──────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000.

.
```

APPENDIX C: AIC Optimal Lag Selection

Lag-order selection criteria

Sample: 1974 thru 2024 Number of obs = 51

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-1099.78				2.7e+11	43.3637	43.4506	43.591
1	-688.731	822.09	36	0.000	113472	28.6561	29.2641	30.247*
2	-630.889	115.68	36	0.000	51085*	27.7995	28.9286*	30.7541
3	-600.16	61.457	36	0.005	73495.3	28.0063	29.6564	32.3245
4	-555.159	90.001*	36	0.000	71647.5	27.6533*	29.8245	33.3351

* optimal lag

APPENDIX D: Jansen Cointegration

Johansen tests for cointegration
Trend: Constant Number of obs = 51
Sample: 1974 thru 2024 Number of lags = 4

Maximum rank	Params	LL	Eigenvalue	Trace statistic	Critical value 5%
0	114	-632.29751	.	154.2763	94.15
1	125	-604.2536	0.66705	98.1885	68.52
2	134	-584.62023	0.53696	58.9218	47.21
3	141	-568.23624	0.47403	26.1538*	29.68
4	146	-559.76744	0.28259	9.2162	15.41
5	149	-555.2069	0.16376	0.0951	3.76
6	150	-555.15934	0.00186		

* selected rank

APPENDIX E: White's test

White's test
H0: Homoskedasticity
Ha: Unrestricted heteroskedasticity

chi2(13) = 7.56
Prob > chi2 = 0.8711

Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
Heteroskedasticity	7.56	13	0.8711
Skewness	5.52	4	0.2382
Kurtosis	2.40	1	0.1215
Total	15.47	18	0.6292

APPENDIX F: Pair-wise Correlation (Multicollinearity check)

	dAnnua~l	dFerti~e	ddiese~e	dExcha~e	dMoney~y
dAnnualrai~l	1.0000				
dFertilize~e	0.0026	1.0000			
ddieselprice	0.0086	0.1898	1.0000		
dExchangeR~e	-0.3749	-0.1471	-0.0450	1.0000	
dMoneySupply	-0.3262	0.2707	0.0599	0.4519	1.0000

APPENDIX G: Lagrange-Multiplier test for Autocorrelation

Lagrange-multiplier test

lag	chi2	df	Prob > chi2
1	47.2350	36	0.09961
2	43.5696	36	0.18045
3	34.9272	36	0.51948
4	28.9421	36	0.79198

H0: no autocorrelation at lag order

APPENDIX H: JB test for Normality

```
. jb resid
Jarque-Bera normality test:  .4331 Chi(2)  .8053
Jarque-Bera test for Ho: normality:

.
. sktest resid
```

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test Adj chi2(2)	Joint test Prob>chi2
resid	51	0.7809	0.6855	0.24	0.8863

APPENDIX I: Granger Causality

Granger causality Wald tests

Equation	Excluded	chi2	df	Prob > chi2
headlineInflation	RainfallVariabi~y	32.82	4	0.000
headlineInflation	FertilizerPrice	3.6826	4	0.451
headlineInflation	dieselprice	32.916	4	0.000
headlineInflation	ExchangeRate	41.321	4	0.000
headlineInflation	MoneySupply	14.558	4	0.006
headlineInflation	ALL	149.33	20	0.000
RainfallVariabi~y	headlineInflation	9.9465	4	0.041
RainfallVariabi~y	FertilizerPrice	8.39	4	0.078
RainfallVariabi~y	dieselprice	19.241	4	0.001
RainfallVariabi~y	ExchangeRate	5.8523	4	0.210
RainfallVariabi~y	MoneySupply	5.3959	4	0.249
RainfallVariabi~y	ALL	69.006	20	0.000
FertilizerPrice	headlineInflation	7.455	4	0.114
FertilizerPrice	RainfallVariabi~y	13.091	4	0.011
FertilizerPrice	dieselprice	9.7331	4	0.045
FertilizerPrice	ExchangeRate	10.249	4	0.036
FertilizerPrice	MoneySupply	.30868	4	0.989
FertilizerPrice	ALL	50.899	20	0.000
dieselprice	headlineInflation	9.4953	4	0.050
dieselprice	RainfallVariabi~y	25.392	4	0.000
dieselprice	FertilizerPrice	4.6326	4	0.327
dieselprice	ExchangeRate	3.8598	4	0.425
dieselprice	MoneySupply	8.1616	4	0.086
dieselprice	ALL	100.16	20	0.000
ExchangeRate	headlineInflation	24.723	4	0.000
ExchangeRate	RainfallVariabi~y	21.297	4	0.000
ExchangeRate	FertilizerPrice	21.693	4	0.000
ExchangeRate	dieselprice	12.075	4	0.017
ExchangeRate	MoneySupply	25.191	4	0.000
ExchangeRate	ALL	87.293	20	0.000
MoneySupply	headlineInflation	10.906	4	0.028
MoneySupply	RainfallVariabi~y	9.1363	4	0.058
MoneySupply	FertilizerPrice	16.351	4	0.003
MoneySupply	dieselprice	5.9215	4	0.205
MoneySupply	ExchangeRate	.25076	4	0.993
MoneySupply	ALL	61.34	20	0.000

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